

From Waste to Value: Thermal Optimization and Incineration Ash Valorization for Sustainable Resource Recovery



Thermophysical Engineering and Development of Incineration Ash-Based Materials

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Background & Research Urgency

Growing Domestic Waste Challenge

- Increasing volumes of non-recyclable domestic waste require safe and effective treatment.
- Thermal treatment offers a potential approach for waste volume reduction and resource recovery.

Need for Controlled Thermal Treatment

- High-temperature and well-controlled combustion is therefore essential for improving the treatment process.

Dual-Chamber Combustion System

- High secondary-chamber temperature and adequate gas residence time support more complete combustion.



Incinerator Machine

Research Methodology



Environmental Performance & Bottom Ash Characterization



Flue Gas Emission Assessment

- Measurement and monitoring of key flue gas emission parameters, including CO, CO₂, NO_x, SO₂, and particulate matter.
- Evaluation of CO concentration as an indicator of combustion completeness.
- Comparison of measured emission parameters with applicable environmental emission standards.
- Identification of operating conditions that may contribute to more efficient and controlled combustion.



Bottom Ash Characterization

- Collection and characterization of bottom ash generated from the combustion process.
- Determination of residual carbon content to assess combustion completeness.
- Analysis of physical and chemical characteristics, including elemental composition and mineral phases.
- Assessment of the potential presence of hazardous elements/contaminants, where applicable.

Potential for Ash Valorization

- Exploration of bottom ash as a potential secondary raw material.
- Material utilization will be determined based on the experimental characterization results.

Expected Sustainability Impact

- Transforming domestic waste from a disposal burden into a potential value-added resource through the integration of thermochemical processing and materials science.